The Silent Thief: Pancytopenia due to Pernicious Anemia

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Background

- Pancytopenia can stem from a variety of causes, including neoplastic conditions, splenic sequestration, nutritional deficiencies, autoimmune processes, and increased cellular destruction seen in disorders such as disseminated intravascular coagulation (DIC) and thrombotic thrombocytopenic purpura (TTP).
- B12 deficiency is relatively common, affecting up to 12% of adults under 60 and nearly 20% of those older.
- Vitamin B12 is a cofactor in DNA methylation and nucleotide synthesis, and thus deficiencies can cause intramedullary hemolysis, megaloblastic anemia and shortened RBC survival.
- In severe cases, intramedullary hemolysis due to B12 deficiency can present with findings of hemolytic anemia (e.g. high LDH, low haptoglobin, schistocytes) due to a phenomenon known as pseudo-thrombotic microangiopathy (pseudo-TMA).
- The most feared complication is subacute combined degeneration (SCD) secondary to increased levels of methylmalonic acid (MMA) which impairs myelin sheath synthesis in the spinal cord.
- High-risk groups for B12 deficiency particularly include individuals on strict vegan or vegetarian diets, as well as those with absorption issues (e.g. pernicious anemia, atrophic gastritis)
- Pernicious anemia, an autoimmune disorder caused by antibodies targeting intrinsic factor or parietal cells, leads to B12 malabsorption. Diagnosis is confirmed by anti-intrinsic factor and anti-parietal cell antibodies, which are 100% specific, though the sensitivity is only 50%, indicating a substantial risk of false negatives.

Case Presentation

A 59-year-old African American female presented with 6 weeks of progressively worsening fatigue, weakness, poor oral intake, and noticeable unintentional weight loss. She denies any hematemesis, hematochezia, melena, hematuria, or prior history of anemia. She had no other known medical problems and was taking no medications and not using any recreational drugs or alcohol. Vital signs on presentation were all within normal limits. Physical exam revealed 3 out of 5 muscle strength of all extremities and hyperpigmentation of the hands and feet bilaterally (Figure 1). Initial lab findings were significant for pancytopenia and macrocytic anemia with a hemoglobin of 6.9 grams/dL, white blood cell count of 1,330 cells/mm^3, absolute neutrophil count of 350 cells/mm^3, platelets of 21,000 cells/mm^3 and a mean corpuscular volume of 114 fL. Peripheral blood smear (PBS) showed 1+ schistocytes, 1+ dacryocytes, 2 + macrocytes and several hypersegmented neutrophils (Figure 2 and Figure 3). She was transfused 1 unit of packed red blood cells in the ED with a modest response in her hemoglobin.

Figure 1 Hyperpigmentation of the hands and feet.

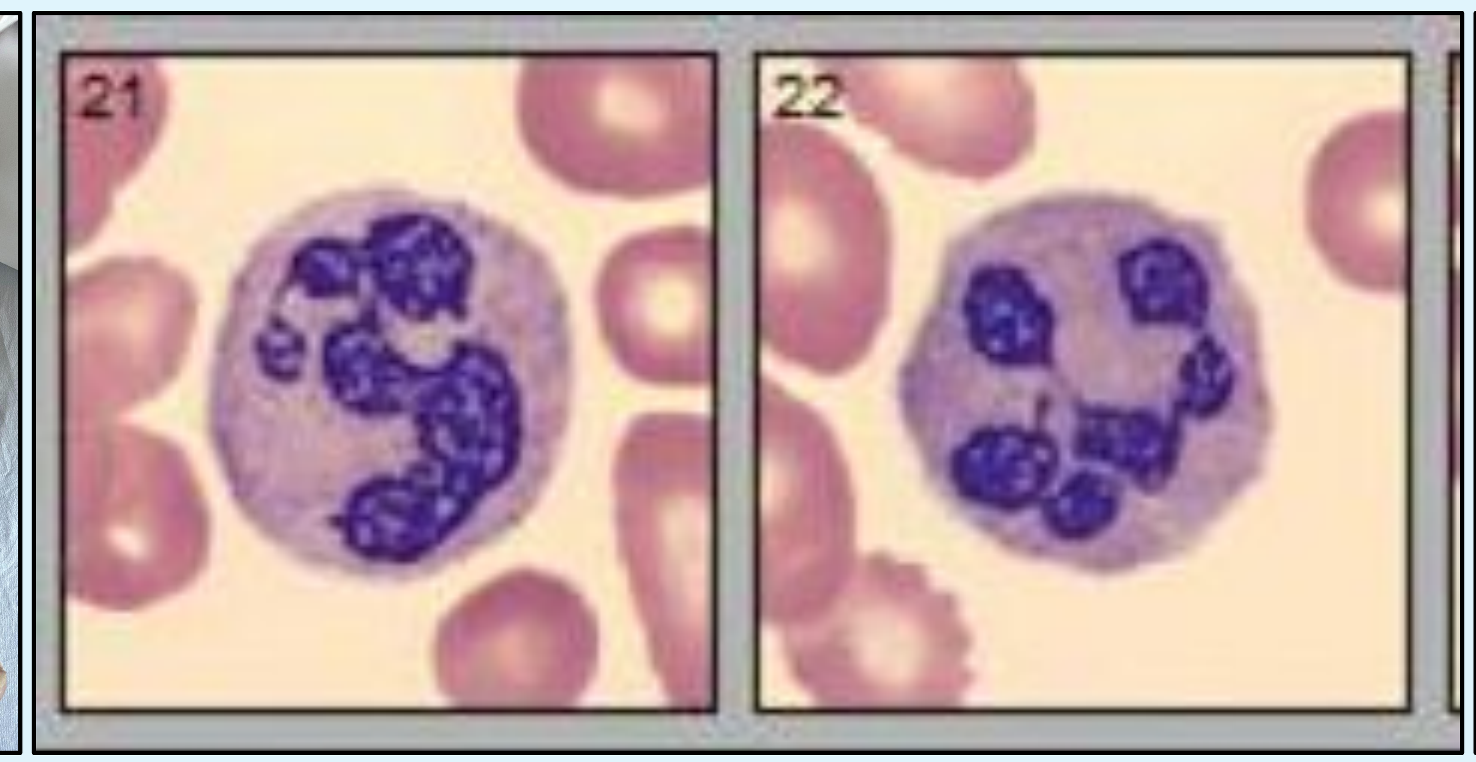

Figure 2 Hypersegmented neutrophils seen on PBS.

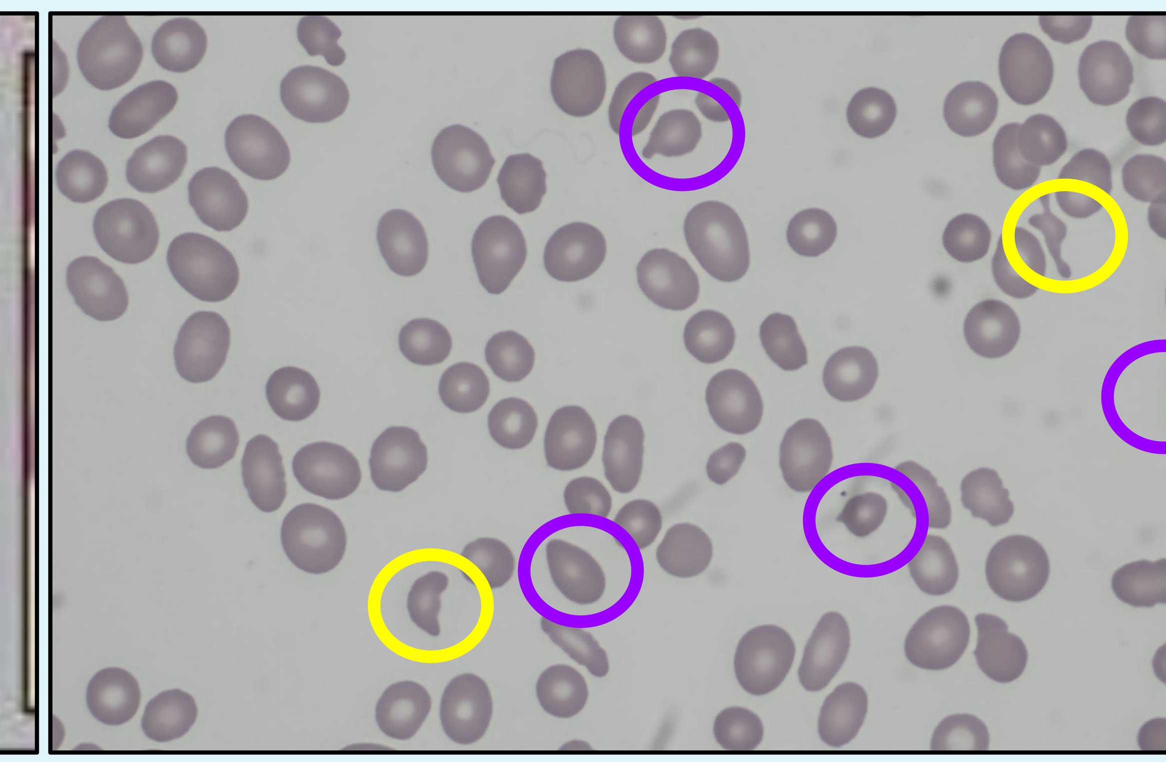

Figure 3 PBS showing schistocytes (circled in yellow) and dacrocytes (circled in purple).

Further evaluation of the pancytopenia included a high LDH (734 units/L), low haptoglobin (<10 milligrams/dL), high CRP (71 units per liter), syphilis/HIV/mono negative, a normal reticulocyte count of 2.2%, a normal folate level of 14.9 nanograms/mL and a very low vitamin B12 level of <150 picograms/mL. A leading diagnosis of pancytopenia secondary to vitamin B12 deficiency was made she was started on on daily subcutaneous 1000mcg vitamin B12 injections, however, the combination of hemolytic markers such as high LDH, low haptoglobin, and thrombocytopenia also raised suspicion for thrombotic thrombocytopenic purpura (TTP). During her hospitalization, a normal ADAMTS13 level excluded TTP and a normal RUQ ultrasound excluded splenic sequestration causes. A complete neurological examination was significant for absent vibratory sense in both feet. A subsequent MRI of the thoracic and cervical spine demonstrated hyperintensity of the dorsal columns consistent with demyelination likely secondary to subacute combined degeneration (Figure 4). A bone marrow biopsy showed a normocellular marrow with moderately severe morphological dyspoiesis excluding most high-grade neoplastic causes (Figure 2). Despite an negative intrinsic factor antibody test, the diagnosis of pernicious anemia was eventually confirmed with an elevated serum gastrin and positive antiparietal cell antibody test. After a week of daily subcutaneous B12 injections, the patient was transitioned to once a week injections, with significant improvement in reticulocyte, ANC, WBC and platelet count, however, anemia persisted (Figure 5).

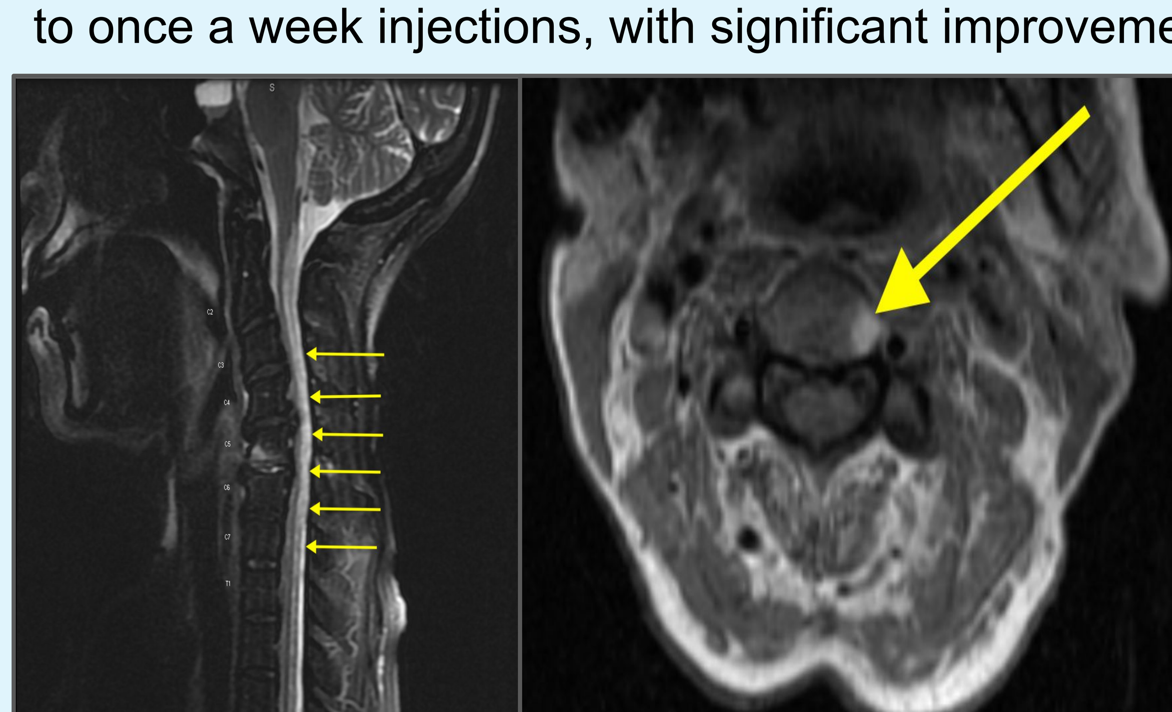

Figure 3 T2 weighted MRI images showing hyperintensity of the dorsal columns (left) and hyperintensity in both the dorsal and lateral columns (right).

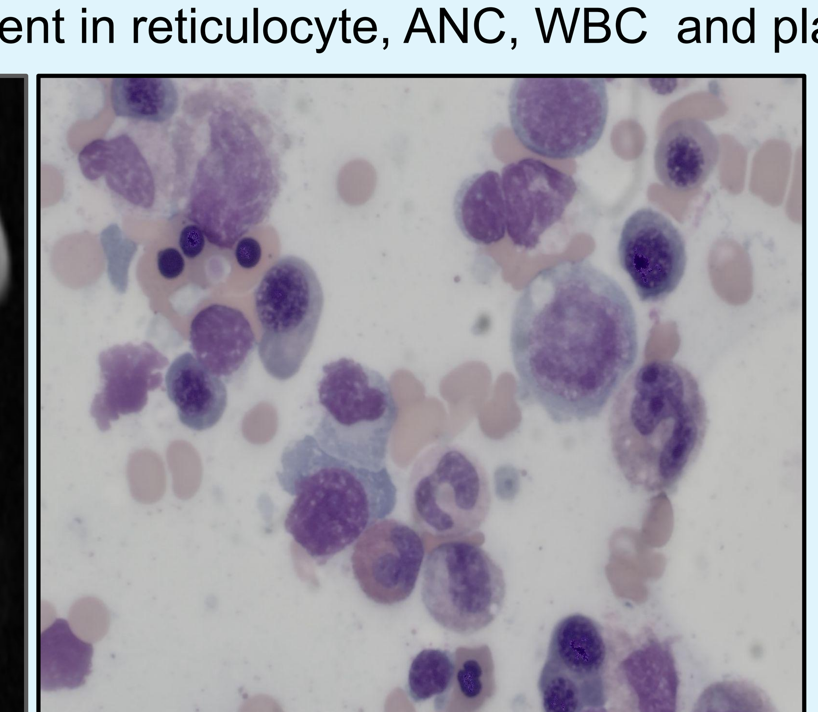

Figure 4 Bone marrow biopsy demonstrating a normocellular marrow with 40-50% cellularity and noticeable dyspoiesis.

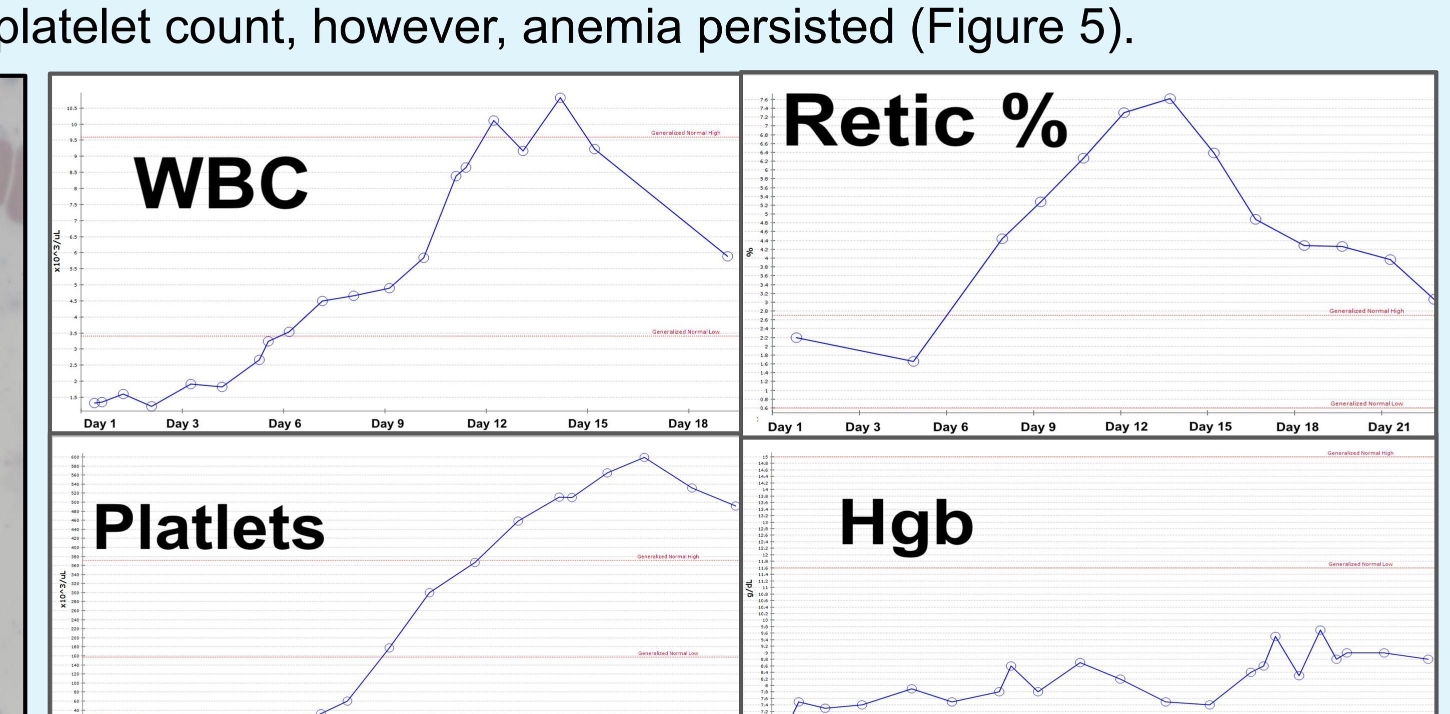

Figure 5 Hematologic response to B12 supplementation over 21 days, demonstrating gradual normalization of all cell counts, except hemoglobin (Hgb) levels, which show a slower yet steady improvement.

Discussion

This case illustrates the complex and diverse manifestations of B12 deficiency. The intramedullary hemolysis that occurs due to deficient DNA methylation causes defective erythropoiesis and results in pseudo-TMA. This phenomenon is characterized by hemolytic anemia, thrombocytopenia, schistocytosis, an elevated LDH and low reticulocyte. Interestingly, bilirubin is only minimally elevated due to the intramedullary destruction of erythrocyte precursors, causing a decrease in hemoglobin. Additionally, the classic finding of hypersegmented neutrophils seen in B12 deficiency was also present. Furthermore, the patients MRI findings and physical exam were significant for the diagnosis SCD, prompting for the rapid supplementation of B12. B12 supplementation typically corrects markers of hemolysis within 1-2 days while reticulocytosis in 3-4 days. While anemia and neurological manifestations may takes weeks to months to resolves as seen in this case. Due to the patient being diagnosed with pernicious anemia, indefinite B12 supplementation will be needed.

Conclusion

This case highlights the critical role of timely B12 supplementation in pernicious anemia to prevent irreversible neurological complications.

Clinical Pearls

1. Severe B12 deficiency can present very similarly to myelodysplastic syndromes (MDS), acute leukemia, and TTP.
2. Hyperpigmentation of the palms and soles is a common manifestation of B12 deficiency due to increased melanin synthesis.
3. When checking for pernicious anemia order **BOTH** anti-intrinsic factor antibodies and antiparietal cell antibodies.
4. If pernicious anemia is suspected, start daily subcutaneous or intramuscular injections of B12 immediately to avoid permanent neurological damage from subacute combined degeneration.

References

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